

Methylsulfonylmethane Induces Cell Cycle Arrest and Apoptosis, and Suppresses the Stemness Potential of HT-29 Cells

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Abstract. *Background/Aim:* Colorectal cancer is one of the most common malignancies worldwide. Small molecule-based chemotherapy is an attractive approach for the chemoprevention and treatment of colorectal cancer. Methylsulfonylmethane (MSM) is a natural organosulfur compound with anticancer properties, as revealed by studies on *in vitro* models of gingival, prostate, lung, hepatic, and breast cancer. However, the molecular mechanisms underlying the effects of MSM in colon cancer cells remain unclear. *Materials and Methods:* Here, we investigated the effects of MSM, especially on the cell cycle arrest and apoptosis, in HT-29 cells. *Results:* MSM suppressed the viability of HT-29 cells by inducing apoptosis and cell cycle arrest at the G₀/G₁ phase. MSM suppressed the sphere-forming ability and expression of stemness markers in HT-29 cells. *Conclusion:* MSM has anti-cancer effects on HT-29 cells, and induces cell cycle arrest and apoptosis, while suppressing the stemness potential.

Colorectal cancer is one of the most commonly diagnosed malignancies worldwide, and its prevalence is increasing in the recent years (1). It is classified on the basis of histopathological features, including the clinical, morphological, and molecular characteristics of the cancer (2).

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Key Words: Methylsulfonylmethane, apoptosis, cell cycle arrest, stemness, HT-29 cells.

Among the various histopathological types, adenocarcinoma accounts for 95% to 98% of all cases of colorectal cancer (3, 4). Colorectal cancer is usually treated by surgery, radiation therapy, or chemotherapy (5). Among these treatment strategies, chemotherapy can generally be administered either before or after surgery for reducing tumor size and the risk of recurrence (6). Chemotherapy may also be administered for providing relief from the symptoms of colorectal cancer in cases where the tumor cannot be removed surgically, or in cases of severe metastasis to other sites of the body (6). Therefore, it is necessary to develop novel chemotherapeutic agents for targeting molecular abnormalities at various stages of malignancies, which requires continuous efforts for the discovery of novel compounds and elucidation of the mechanisms underlying their mode of action.

Methylsulfonylmethane (MSM) is a simple organosulfur compound, which is also known as dimethylsulfone, methylsulfone, sulfonyl bis methane, organic sulfur, or crystalline dimethyl sulfoxide (7). MSM occurs naturally in plants, animals, and various natural products (7). It has been recognized as a safe substance, and the US Food and Drug Administration (FDA) classified MSM as an 'Generally Recognized as Safe (GRAS)' grade molecule (8). It has been reported that MSM has numerous physiological functions, and is particularly well-known due to its analgesic properties in chronic pain caused by inflammatory diseases (9).

In the recent past, numerous studies have reported the anticancer properties of MSM using *in vitro* models (10). For instance, it has been reported that MSM induces ER stress-mediated apoptosis in HCT116 cells (11). Another study demonstrated the antiproliferative effects of MSM and suggested its inhibitory potential in antagonizing the invasive and migratory ability of prostate cancer cells (12). MSM induces apoptosis in hepatic cells, including the HepG2 and Huh7 cells, via the extrinsic apoptotic pathway (13). Recently, our group also demonstrated the anticancer effect of MSM in YD-38 gingival cancer cells and A549 lung

cancer cells and specially emphasized on the MSM-induced changes in cell viability, cell cycle, and apoptosis of cancer cells (14, 15). Our study additionally elucidated the mechanisms underlying the effect of MSM on the proliferation and invasiveness of breast cancer cells, and the changes in the oncogenic signal transducer and activator of transcription 3 (STAT3) and STAT5b signaling (16). However, the effects and possible mechanisms underlying the anticancer properties of MSM in HT2-9 colorectal cancer cells are yet to be fully explored.

In this study, we investigated the molecular mechanisms underlying the inhibitory effects of MSM in HT-29 cells, a typical model of colorectal adenocarcinoma. We evaluated the effects of MSM on cell viability, cell cycle, and the metastatic potential of HT-29 cells, including the invasiveness and sphere-forming ability.

Materials and Methods

Reagents. The stock solution of MSM (1 M) (Sigma-Aldrich, St. Louis, MO, USA) was prepared in Roswell Park Memorial Institute (RPMI)-1640 medium (Gibco, Grand Island, NY, USA) and stored in the dark at 4°C. It was immediately diluted in RPMI-1640 medium before use. Reagents for 3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide (MTT), propidium iodide (PI) were supplied from Sigma-Aldrich (Sigma-Aldrich, St. Louis, MO, USA). Antibodies specific to cleaved poly-ADP ribose polymerase (PARP), cleaved caspase 9, B-cell lymphoma 2 (Bcl-2), B-cell lymphoma-extra-large (Bcl-XL) and bcl-2-like protein 4 (Bax) were supplied from Cell Signaling Technology (Cell Signaling Technology, Danvers, MA, USA). Antibodies specific to cyclin D, cyclin E, CDK4, Rb, phospho-Rb and Actin were supplied from Santa Cruz Biotechnology (Santa Cruz, Delaware Avenue, CA, USA). The anti-rabbit IgG horseradish peroxidase (HRP) conjugated secondary antibody and anti-mouse IgG HRP-conjugated secondary antibody were purchased by Merck (Merck, Darmstadt, Germany). 3,3'-Diethyloxycarbocyanine iodide (DiOC6) was supplied from Enzo (Enzo, NY, USA). The FITC-Annexin V apoptosis detection kit I was supplied from BD Biosciences (BD Biosciences, San Jose, CA, USA).

Cell culture. All the experiments were conducted by following the same methodology we used in our previously published work (17). Human colorectal adenocarcinoma cell line (HT29) was purchased from the American Type Culture Collection (Manassas, MD, USA). Cells were cultured and sub-passaged in RPMI 1640 (Gibco, Grand Island, NY, USA) supplemented with 10% fetal bovine serum (FBS; Invitrogen, Carlsbad, CA, USA) and 1% antibiotic-antimycotic (ABAM; Invitrogen). Cells were cultured in a humidified incubator with 5% CO₂ at 37°C. Routine culture media renewals were made every 3 times a week.

Cell viability assay. MTT assay was conducted to measure the viability of cells. Approximately 1×10⁴ cells/100 µl medium cells were seeded in 96-well culture plates and were cultured overnight. After 20 h of incubation, various concentrations of MSM were added to the wells and incubated further for 24 h and 48 h. Then, MTT (5 mg/ml) was added to each well and incubated for 4 h at

37°C. The formazan product was dissolved by adding 100 µl of dimethyl sulfoxide (DMSO) to each well, and the absorbance at 570 nm was measured using an Ultra Multifunctional Microplate Reader (TECAN, Durham, NC, USA). All measurements were performed in triplicates and repeated at least three times.

Cell cycle analysis. Cell cycle was measured by PI staining and flow cytometry analysis. Approximately 1.5×10⁵ cells/ml of HT29 cells were seeded into 60 mm culture plates and incubated 24 h. Cells were treated with different concentrations of MSM further 24 h. The Cells were washed, harvested with trypsin-EDTA and fixed with 70% ethanol at 20°C. After fixation, cells were washed twice with cold PBS and centrifuged to discard the supernatant. The pellet was resuspended with PBS and stained with PBS containing 50 µg/ml PI and 100 µg/ml RNase A for 30 min in the dark. DNA contents data were analyzed using a FACSCalibur instrument and CellQuest software (BD Biosciences, San Jose, CA, USA).

Annexin V and PI staining. HT29 cells (1.5×10⁵ cells/ml) were seeded in a 60 mm plates and incubated for 24 h. Next, the cells were treated with different concentrations of MSM and were incubated for 24 h. Then, cells were harvested and collected. Annexin V and PI staining were performed using FITC-Annexin V apoptosis detection kit I (BD Biosciences, San Jose, CA, USA). Test procedure was following the manufacturer's instructions. The proportion of apoptotic cells was determined flow cytometry analysis with an FACSCalibur instrument and CellQuest software (BD Biosciences, San Jose, CA, USA).

Analysis of mitochondrial membrane potential. HT29 cells (2.5×10⁵ cells/well) were cultured in 60 mm culture dishes for 24 h and treated with each concentration of MSM. Cells were harvested and transferred to 1.5 ml tubes. DiOC6 (0.1 M) was added to the each well and incubated for 15 min in the dark at 37°C. Then, cells were centrifuged (300 × g, 5 min, 4°C), washed twice with PBS and resuspended in 200 µl of PBS. Data were analyzed by FACSCalibur instrument and CellQuest software (BD Biosciences). The entire protocol was conducted in minimal light.

Western blot analysis. Cells were harvested and lysed with buffer 50 mM Tris (pH 7.4), 1.5 M sodium chloride, 1 mM EDTA, 1% NP-40, 0.25% sodium deoxycholate, 0.1% sodium dodecyl sulfate (SDS), and a protease inhibitor cocktail. Then, lysates were incubated on ice for 1 h and centrifuged at 17,000 × g for 30 min at 4°C. Protein content was quantified using Bradford assay (Bio-Rad, Hercules, CA, USA). Lysate were boiled at 100°C for 5 min. The protein extract were separated by using 10-12% sodium dodecyl sulfate polyacrylamide gel electrophoresis (SDS-PAGE) and transferred to polyvinylidene difluoride membranes (PVDF; Millipore, Billerica, MA, USA), which were blocked in 5% non-fat dried milk dissolved in Tris buffered saline containing Tween-20 (2.7 M NaCl, 53.65 mM KCl, 1 M Tris-HCl, pH 7.4, and 0.1% Tween-20) for 1 h at room temperature. Then, the membranes were incubated overnight at 4°C with specific primary antibodies. Membranes were then washed with Tris-buffered saline containing 0.05% Tween 20 and incubated with the secondary antibodies (HRP conjugated anti-rabbit or anti-mouse IgG) for 1 h at room temperature. After washing, Specific protein bands were detected using West-Zol Plus and western blot detection system (iNtRON Biotechnology, Sungnam, Republic of Korea), according to the

Table I. RT-PCR primer sequences, annealing temperature and product sizes.

Gene	Sequence (5' to 3')	Annealing T _m (°C)	Product size (bp)
<i>SOX2</i>	Forward: GTGGAAACTTTTGTCTGGAGA Reverse: GGCAGCGTGTACTTATCCTT	56	150
<i>OCT4</i>	Forward: CACGTAGGTTCTTGAATCCC Reverse: CACATCGGCCTGTGTATATC	56	185
<i>NANOG</i>	Forward: AAAAGGAAGACAAGGTCCCA Reverse: CCTGTTTGTAGCTGAGGTTC	56	159
<i>GAPDH</i>	Forward: CAACTACATGGCTGAGAACG Reverse: GATGATGACCCTTTTGGCTC	56	190

manufacturer's instructions. Relative densitometric quantification of western blot data was performed using Image J software and normalized to Actin.

Tumor sphere formation assay. To generate spheroid bodies, HT29 cells (5×10^3 cells/well) were seeded in DMEM/F12 media containing EGF, bFGF and B27 supplements in low attachment plates at 37°C/5% CO₂. After 24 h incubation, cells were treated with different concentration of MSM. The treatment day was considered as day 0 and images were taken at days 0, 3, and 7 using microscope. Total RNA was isolated using TRizol reagent from the sphere at day 7 and analyzed using RT-PCR. The number of spheres was counted with a microscope at day 7.

RNA extraction and cDNA synthesis. Total RNA was extracted from HT29 cells using TRizol (Invitrogen, Karlsruhe, Germany) reagents. Cell were harvested at 1.5 ml tubes and resuspended with 1 ml of TRizol reagents. Two hundred µl of chloroform was added and the mixture was shaken for 10 s, maintained at 4°C for 5 min, and then centrifuged at 4°C for 15 min. The supernatant was transferred to a new test tube and mixed with an equal amount of isopropanol. Then, the mixture was maintained at 4°C for 15 min to collect RNA pellets. Isopropanol was removed from the solution by centrifuging at 4°C for 15 min, and then the pellet was washed with 75% ethanol. The mRNA pellet was dissolved in RNase-free water. The purity of the extracted RNA was confirmed using a spectrophotometer (ND-100, Nanodrop Technologies Inc., Wilmington, DE, USA), and only RNA with a purity (OD value of 230 nm/260 nm) greater than 1.8 was selected and stored at -70°C until the experiment was conducted. To synthesize cDNA, 1 µg of RNA and 1 µl each of oligo-dT (Invitrogen, MA) and RNase-Free Water were added. The RNA was incubated at 80°C for 3 min, and cDNA was synthesized using 4 µl of 5 × RT (reverse transcription) buffer, 5 µl of 2 mM dNTPs, 0.5 µl of RNase inhibitor (Promega, Madison, WI, USA), and 1 µl of M-MLV (Moloney-murine leukemia virus) RT (Promega).

Reverse transcription polymerase chain reaction (RT-PCR). Gene sequence information was retrieved from database from NCBI nucleotide and the Ensembl Genome Browser. PRIMER3 software were utilized for synthesize the primers for amplification of the genes (Table I). RT-PCR was performed using a C1000 Thermal Cycler (Bio Rad, Hercules, CA, USA). The condition of RT-PCR reaction was constructed as follows: 2 µl diluted cDNA (50 ng/µl)

was added to 16 µl reaction premix (Bio Rad) and 1 µl (5 pmol/µl) each forward and reverse primers. The RT-PCR was conducted as follows: initial denaturation at 94°C for 10 min followed by 35 cycles of denaturation at 94°C for 30 s, annealing at 60°C for 30 s, and extension at 72°C for 30 s. GAPDH was used as reference gene.

Statistical analysis. All data were expressed as mean ± standard deviation from three independent experiments. Graph Pad Prism software (Graph Pad Software, La Jolla, CA, USA) was used to evaluate the data. The statistical significance (* $p < 0.05$, ** $p < 0.01$, or *** $p < 0.001$) was assessed by paired student's t-test and one-way analysis of variance (ANOVA) followed by post-hoc comparison (Tukey's HSD).

Results

MSM suppresses the viability of HT-29 cells in a dose-dependent manner. The cytotoxicity of MSM in HT-29 cells was measured by the MTT assay. The cells were prepared and treated with MSM at the concentrations of 0, 50, 100, 150, 200, 250, 300, or 350 µM for 24 or 48 h. The results demonstrated that the viability of HT-29 cells was suppressed by MSM in a dose-dependent manner at 24 h (Figure 1A). At 24 h, the half maximal inhibitory concentration (IC₅₀ value) was approximately 200 mM in HT-29 cells (Figure 1A). A similar trend was observed for the group that was treated with MSM for 48 h, however, the effect was more severe compared to that of the group treated with MSM for 24 h (Figure 1B). The subsequent experiments were conducted on the basis of these results.

MSM induces cell cycle arrest at the G₀/G₁ phase and increases the sub-G₁ population of HT-29 cells. Induction of cell cycle arrest is an important goal for the chemoprevention and treatment of abnormally-activated cells in cancer (18). In particular, the apoptotic cells in the sub-G₁ phase contain hypodiploid fragmented DNA (19). Therefore, in order to identify the effects of MSM on the cell cycle of HT-29 cells, we examined the percentage of cells in the different phases of the cell cycle using FACS. We also determined the protein levels of cell cycle-related proteins, including cyclin

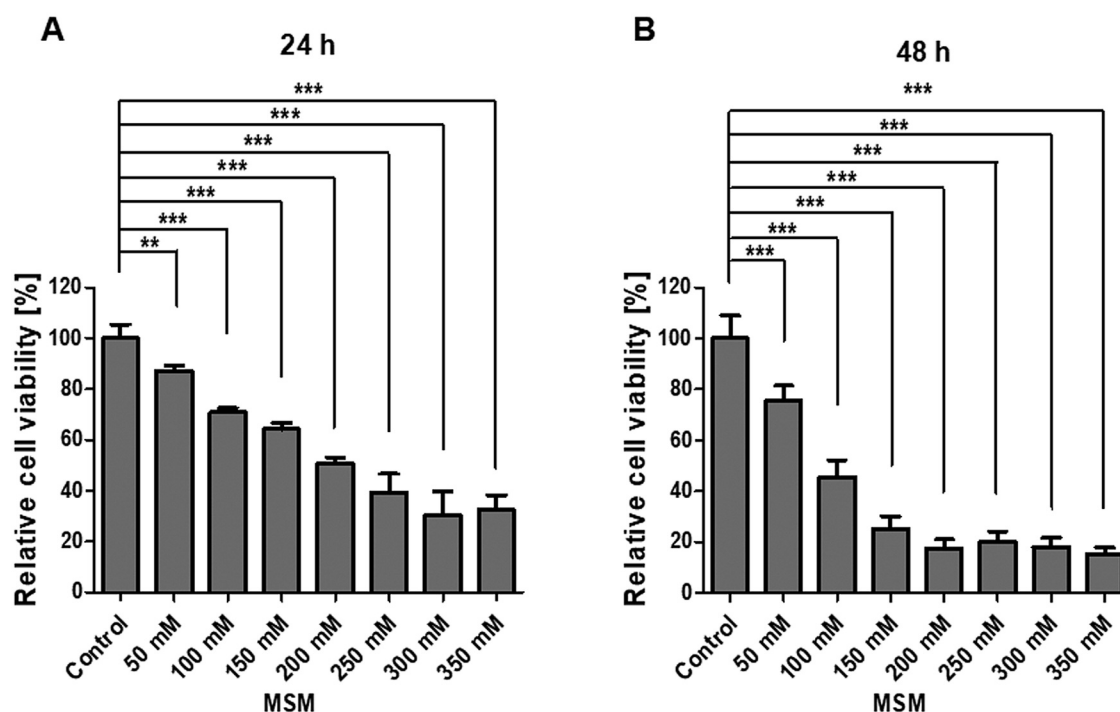


Figure 1. Effects of MSM on the viability of HT-29 colon adenocarcinoma cells. The cells were treated with MSM at various concentrations of 0, 50, 100, 150, 200, 250, 300, or 350 mM for (A) 24 h or (B) 48 h, and cell viability was assessed by the MTT assay. Cell viability is represented as the percent relative absorbance (OD570) of the treated group compared to that of the control group. The data are expressed as the mean $\pm$ SD of three independent experiments. The asterisks denote statistical significance (one-way ANOVA, ** p <0.01, *** p <0.001) of the post-hoc comparisons (Tukey's HSD) with that of the control, for each concentration of MSM.

dependent kinase 4 (CDK4), cyclin E, cyclin D, retinoblastoma (Rb), and phospho Rb (p-Rb), in HT-29 cells. The results demonstrated that when the HT-29 cells were treated with MSM for 24 h, the ratio of cells in the G₀/G₁ phases increased in a concentration-dependent manner, and treatment with MSM increased the population of cells in the sub-G₁ phase, indicative of chromatin fragmentation (Figure 2A, B). Additionally, the results of western blotting indicated that treatment with MSM decreased the expression of cyclin D, cyclin E, CDK4, and p-Rb proteins (Figure 2C). These results indicated that MSM induced cell cycle arrest at the G₀/G₁ phase and increased cell death of HT-29 cells.

MSM induces apoptosis of HT-29 cells. In order to identify the mechanism underlying MSM-induced apoptosis in HT-29 cells, we determined the exposure of phosphatidylserine on the cell surface, and the increase in nuclear membrane permeability, which are important characteristics of apoptosis, using Annexin V fluorescein isothiocyanate (FITC)/propidium iodide (PI) staining, that is commonly used for detecting apoptosis (20). The results demonstrated that treatment with MSM increased the population of annexin V⁺/PI⁻ (early apoptotic) and annexin V⁺/PI⁺ (late apoptotic) cells (Figure

3A, B). Subsequently, the protein levels of the genes related to apoptosis, including cleaved caspase 9 and cleaved Poly (ADP-ribose) polymerase (PARP), were evaluated by western blotting. As depicted in Figure 3C, the protein levels of cleaved PARP and cleaved caspase 9 increased following treatment with MSM. Altogether, these results indicated that MSM induced apoptosis of HT-29 cells in a concentration-dependent manner.

Mitochondria are involved in the MSM-induced apoptosis of HT-29 cells. Mitochondrial outer membrane permeabilization (MOMP) is a major factor in the intrinsic apoptotic pathway (21). MOMP is affected by changes in the expression levels of genes in the B-cell lymphoma 2 (Bcl-2) family, and triggers the caspase cascade by inducing the cytosolic localization of cytochrome *c* (21). In this study, we aimed to identify the mechanism underlying the apoptosis induced by MSM. For this, we examined the changes in the protein levels of Bcl-2, Bcl-XL, and Bax and the mitochondrial membrane potential following treatment with MSM, using DiOC6 staining. As depicted in Figure 4A, the results of western blotting indicated that treatment with MSM suppressed the expression of Bcl-2 and Bcl-XL proteins but

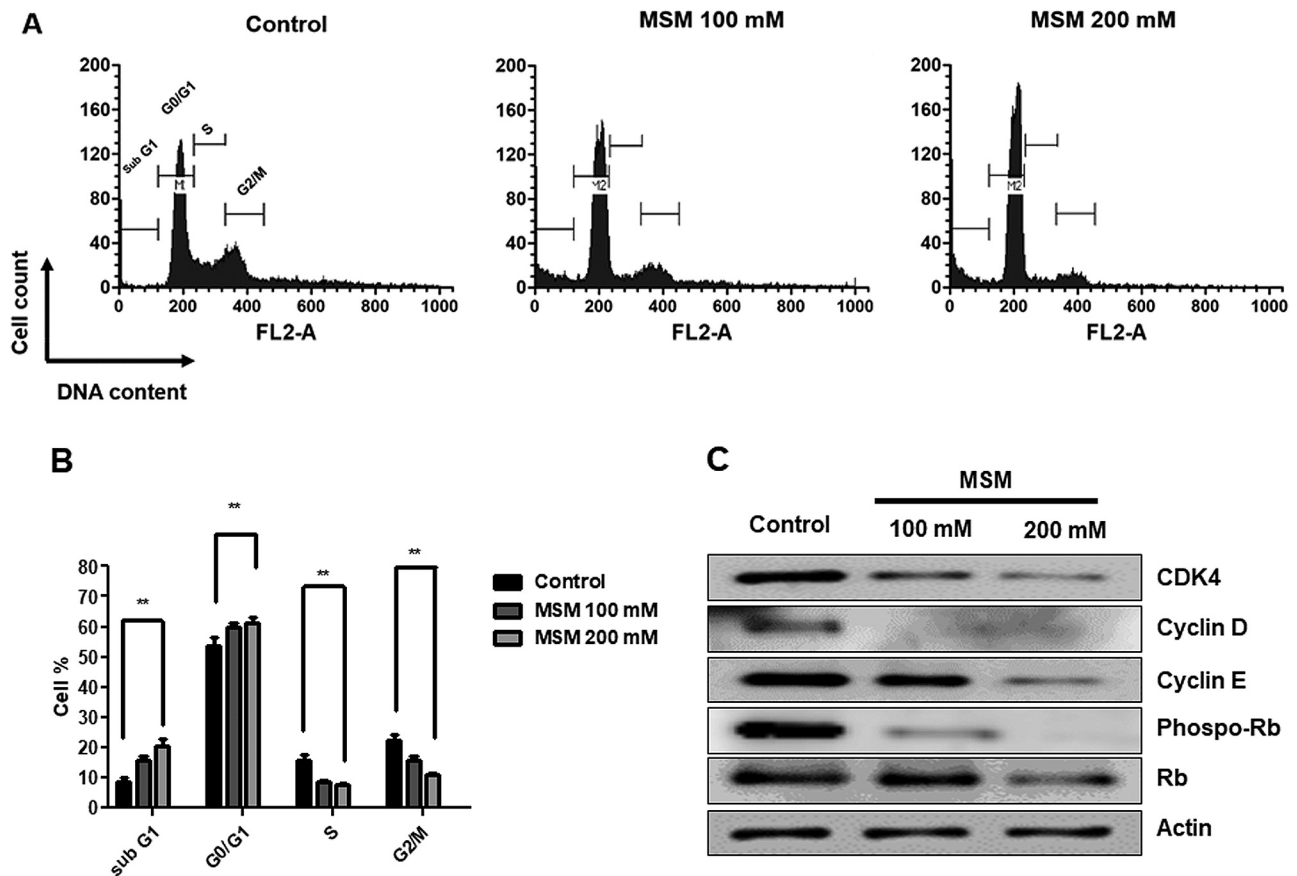


Figure 2. MSM induces cell cycle arrest and modulates the expression of cell cycle regulators in HT-29 cells. (A) Results of flow cytometry analysis showing the changes in the cell cycle composition of HT-29 cells following treatment with MSM. (B) Graphical representation of the relative cell cycle composition of HT-29 cells. The data are expressed as the mean $\pm$ SD of three independent experiments. The asterisks denote statistical significance (** $p < 0.01$) of post-hoc comparisons (Tukey's HSD) with the control, for each concentration of MSM. (C) The expression levels of cyclin dependent kinase 4 (CDK4), cyclin D, cyclin E, p53, retinoblastoma (Rb), and phospho-Rb (p-Rb) in HT-29 cells treated with MSM. The protein levels were examined by western blotting. Actin was used as the internal control for sample loading.

induced the expression of Bax protein. Furthermore, the mitochondrial membrane potential of HT-29 cells decreased following treatment with MSM (Figure 4B, C). These results demonstrated that mitochondrial disruption is involved in MSM-induced apoptosis.

MSM suppresses the stemness potential of HT-29 cells. In order to evaluate the effect of MSM on the stemness potential of HT-29 cells, we conducted a sphere-forming assay. In our study, sphere formation was observed after 0, 3, or 7 days of incubation in an optimized sphere-forming medium on ultra-low attachment dishes. The number of spheres was counted after 7 days of incubation, with a microscope. We additionally examined the relative mRNA levels of the stemness markers, including Sex determining region Y-box 2 (SOX2), octamer-binding transcription factor 4 (OCT4), and NANOG. The results indicated a reduction in

the size and number of the spheres in the MSM-treated group, compared to those of the control group (Figure 5A, B). Furthermore, MSM reduced the mRNA levels of SOX2, OCT4, and NANOG in a concentration-dependent manner (Figure 5C). These results indicated that MSM suppressed the stemness potential of HT-29 cells.

Discussion

Natural phytochemicals or synthetically-derived small molecules are attractive subjects of recent studies, for the prevention, cure, or delay of cancer progression in the field of "Cancer chemoprevention with dietary phytochemicals" (22). MSM is a small organosulfur compound with antioxidant and anti-inflammatory properties (10). Numerous studies have recently demonstrated the anticancer properties of MSM (10). In this study, we examined the effects of

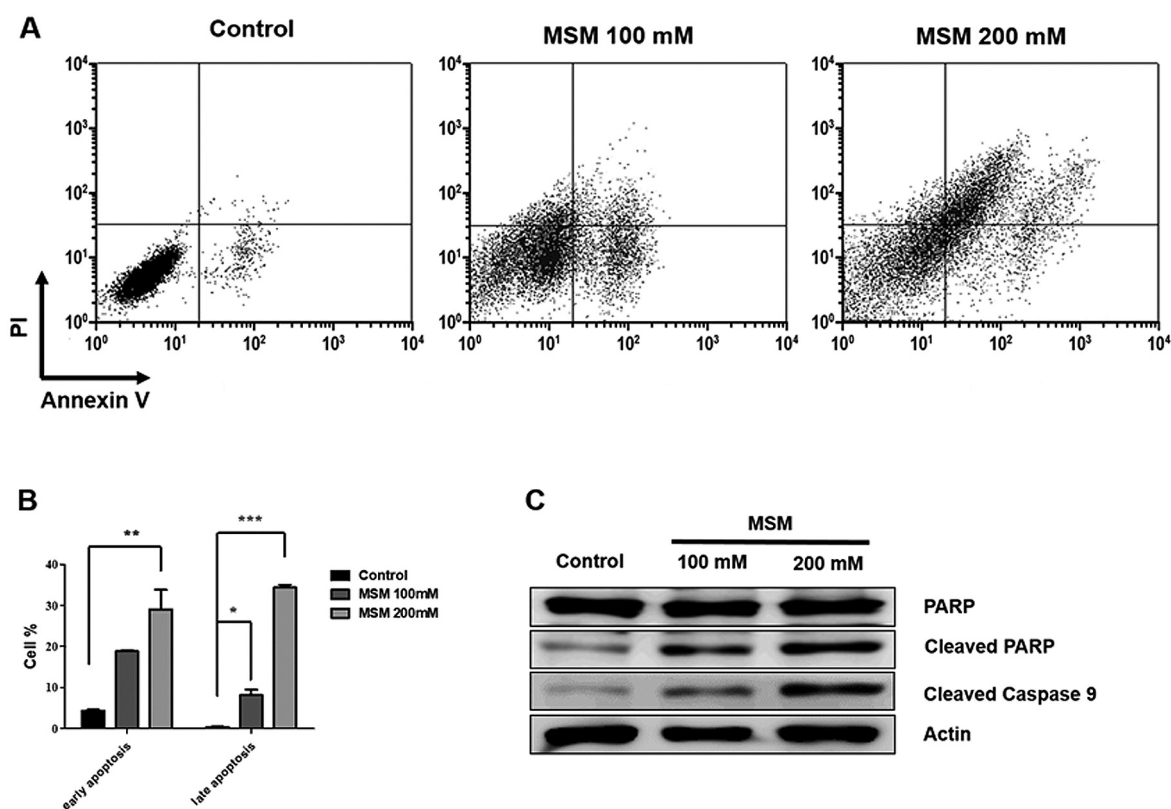


Figure 3. MSM induces the caspase-mediated apoptosis of HT-29 cells. (A) Flow cytometry using Annexin V fluorescein isothiocyanate (FITC)/propidium iodide (PI) staining. (B) Graphical representation of the relative percentage of apoptotic cells. The data are expressed as the mean $\pm$ SD of three independent experiments. The asterisks denote statistical significance (one-way ANOVA, * p <0.05, ** p <0.01, *** p <0.001) of post-hoc comparisons (Tukey's HSD) with the control, for each concentration of MSM. (C) The protein levels of PARP, cleaved PARP, and cleaved caspase 9 as analysed by western blotting. Actin was used as the internal control for sample loading.

MSM on the cell cycle and apoptosis of HT-29 cells. The effect of MSM on the stemness potential of HT-29 cells was also evaluated.

We first attempted to investigate the inhibitory effect of MSM on colon cancer cells. Treatment with MSM decreased the viability of H29 colon cancer cells in a dose-dependent manner (Figure 1A, B). The IC₅₀ value was approximately 200 mM (Figure 1A, B) after 24 h of treatment with MSM, and the result was similar to that of previous studies (15). It can be estimated that the decrease in cell viability following treatment with MSM was related to cell cycle inhibition and apoptosis.

It has been previously demonstrated that MSM induces cell cycle arrest at the G₀/G₁ phase in prostate and gingival cancer cells (15). It has been additionally reported that the cell cycle regulatory genes, including the CDKs and cyclins, are regulated by MSM (15). Although cancer cells have different gene expression profiles and carry different mutations depending on the type of cancer cell, the results of our study demonstrated that MSM induces cell cycle arrest at the G₀/G₁

phase in HT-29 cells (Figure 2A, B). We additionally observed that the expression levels of CDK4, cyclin D, and cyclin E proteins are reduced following treatment with MSM (Figure 2C). We also observed that the phosphorylation of RB was inhibited by MSM (Figure 2C). The results of our study are similar to those of previous studies.

The processes of cell death are still being investigated, and various mechanisms are being reported to be associated with cell death (23). Among these, the process of apoptosis has been well studied, and is an attractive target for the development of anticancer agents (24). Recent studies have reported that MSM induces apoptosis in lung, hepatic, gingival, and breast cancer cells (12, 14, 15). Apoptotic cells have specific characteristics in several respects (25). In particular, the changes in the permeability of the nuclear membrane and surface exposure of phosphatidylserine are considered to be some of the most important features of apoptotic cells (20). In this study, treatment with MSM increased the population of cells in the sub-G₁ phase, which was indicative of chromatin degradation. Additionally, the

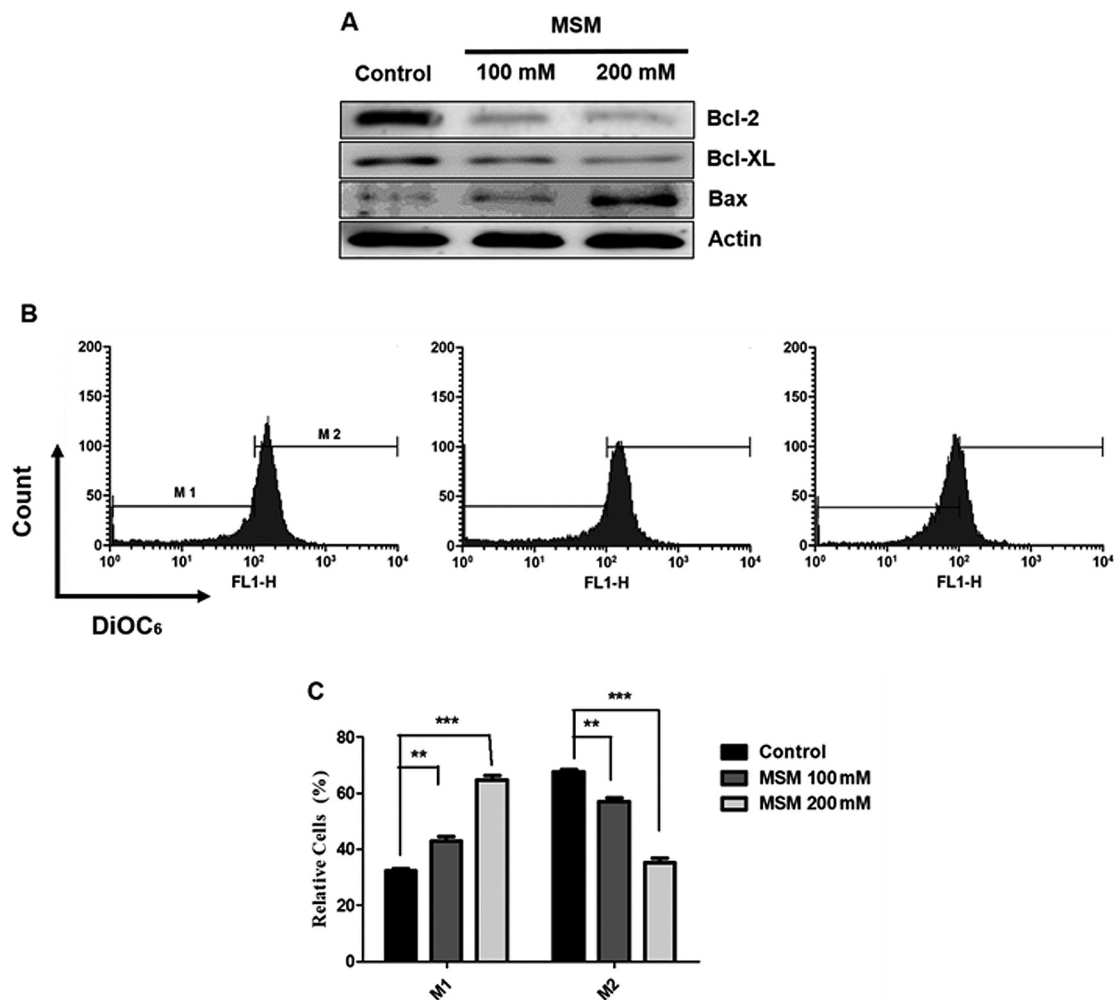


Figure 4. The involvement of mitochondria in the process of MSM-induced apoptosis in HT-29 cells. (A) The expression levels of Bcl-2, Bcl-XL, and Bax were measured by western blotting. Actin was used as the internal control for sample loading. (B) The relative mitochondrial membrane potential of the HT-29 cells. The HT-29 cells were incubated with 300 mM MSM for 24 h and stained with DiOC₆. The data were obtained by flow cytometry. The point for separating the cell population was located at FL1-H 102, which divided the population of HT-29 cells into M1 and M2. (C) Graphical representation of the percentage of M1 and M2 cells. The data are expressed as the mean $\pm$ SD of three independent experiments. The asterisks denote statistical significance (one-way ANOVA, ** $p < 0.01$, *** $p < 0.001$).

results of the Annexin V/PI staining assay revealed that the population of HT-29 cells in early and late apoptosis increased after treatment with MSM (Figure 3A, B).

The apoptotic pathway is categorized into the extrinsic apoptotic pathway and the intrinsic apoptotic pathway (26). Both the intrinsic and extrinsic pathways eventually induce cell death by inducing the caspase cascade and the expression of cleaved PARP (26). In this study, the expression levels of cleaved caspase 9 and cleaved PARP increased following treatment with MSM (Figure 3C). Additionally, the results of FACS revealed that treatment with MSM decreased the mitochondrial membrane potential and changed the expression of Bcl-2 genes (Figure 4A, B).

These results indicated that the mitochondrial pathway was involved in the MSM-induced apoptosis of HT-29 cells.

Then, we specifically investigated the effect of MSM on the stemness potential of HT-29 cells. The cancer stem cell (CSC) theory defines CSCs as small groups of cells within the tumour that are responsible for tumor initiation, growth, and recurrence (27). In particular, the potential for self-renewal and differentiation is a key feature of CSCs (28). The sphere-forming ability and expression of stem cell markers are the characteristic features used for assessing the stemness potential of cancer cells (28). In order to investigate the effect of MSM on the stemness potential of HT-29 cells, we performed sphere-forming assays for analyzing the expression

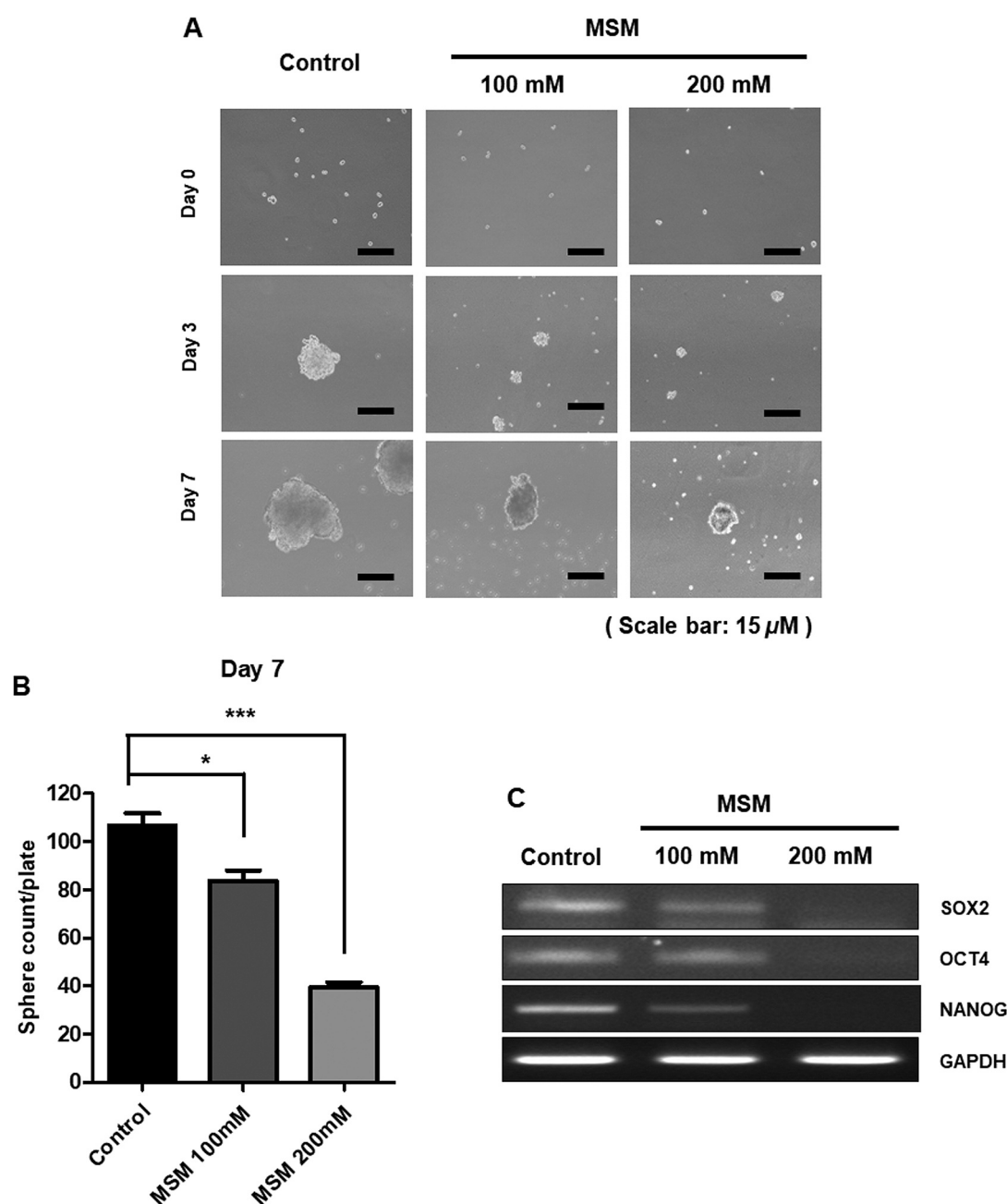


Figure 5. MSM suppresses the stemness potential of HT-29 cells. (A) The effects of MSM on the sphere-forming ability of HT-29 cells. Representative photographs of the cells (scale bar, 15 μ m). The cells were incubated in sphere-forming media for 7 days with different concentrations of MSM. (B) The effects of MSM on the number of spheres of HT-29 cells after 7 days incubation. The data are expressed as the mean $\pm$ SD of three independent experiments. The asterisks denote statistical significance (one-way ANOVA, * p <0.05, *** p <0.001) of post-hoc comparisons (Tukey's HSD) with the control, for each concentration of MSM. (C) Effect of MSM on the mRNA expression of SOX2, OCT4, NANOG, and GAPDH as analysed by RT-PCR. GAPDH was used as the internal control for sample loading.

levels of the stemness marker genes. The results of our study demonstrated that treatment with MSM reduced the size and number of spheres of HT-29 cells (Figure 5A, B). Furthermore, the expression of the stemness marker genes,

including SOX2, OCT4, and NANOG, was suppressed by MSM (Figure 5C). To the best of our knowledge, this study is the first to demonstrate the inhibitory effect of MSM on the stemness potential of colon cancer cells.

In conclusion, this study demonstrated that treatment with MSM can inhibit the viability of HT-29 colon cancer cells by triggering cell cycle arrest and apoptosis and suppressing the stemness potential of HT-29 cells.

Conflicts of Interest

The Authors declare no conflicts of interest regarding this study.

Authors' Contributions

Y.M.Y. and K-J.J. designed the experiments. D.H.K., D.Y.K., and N.S. conducted most of the experiments. E.S.J. and A.R. helped in some experiments. Y.M.Y., K-J.J., D.H.K., D.Y.K., and N.S. evaluated the data, and D.H.K. and K-J.J. wrote the manuscript. All Authors helped to revise the article and accepted the final version for publication.

Acknowledgements

This work was supported by the National Research Foundation of Korea (NRF) grant funded by the Korean government (MSIT) (No. 2018R1C1B6006146) and by Basic Science Research Program through the National Research Foundation of Korea (NRF) funded by the Ministry of Education (No. 2018R1D1A1B07048651 and 2020R1I1A2073517).

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Received July 9, 2020

Revised July 25, 2020

Accepted July 31, 2020